Types of Numbers

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Types of Number

You will come across vocabulary such as

- **Integers** and **natural** numbers
- **Rational** and **irrational** numbers
- **Multiples**
- **Factors**
- **Prime** numbers
- **Squares**, **cubes** and **roots**
- **Reciprocals**

Knowing what each of these terms mean is essential.

What are integers and natural numbers?

- **Integers** are **whole** numbers;
  - They can be **positive, negative** and **zero**
  - For example, -3, -2, -1, 0, 1, 2, 3 are all integers
- **Natural** numbers are the **positive integers**
  - They can be thought of as counting numbers
  - 0, 1, 2, 3, 4, ... are the natural numbers
    - Notice that 0 **is** included

What are multiples?

- A **multiple** is a number which can be divided by another number, without leaving a remainder
  - For example, 12 is a multiple of 3
    - 12 divided by 3 is exactly 4
- A **common multiple** is multiple that is shared by more than one number
  - For example, 12 is a common multiple of 4 and 6
- **Even** numbers (2, 4, 6, 8, 10, ...) are multiples of 2
- **Odd** numbers (1, 3, 5, 7, 9, ...) are **not** multiples of 2
- Multiples can be **algebraic**
  - For example, the multiples of k would be $k, 2k, 3k, 4k, 5k. \ldots$

What are factors?

- A **factor** of a given number is a value that divides the given number exactly, with **no remainder**
  - 6 is a factor of 18

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- A **common factor** is a factor that is shared by more than one number
  - For example, 3 is a common factor of both 21 and 18

How do I find factors?

- Finding all the factors of a particular value can be done by finding **factor pairs**
- For example when finding the factors of 18
  - 1 and 18 will be the first factor pair
  - Divide by 2, 3, 4 and so on to test if they are factors
    - $18 \div 2 = 9$, so 9 and 2 are factors
    - $18 \div 3 = 6$, so 6 and 3 are factors
    - $18 \div 4 = 4.5$, so 4 is not a factor
    - $18 \div 5 = 3.6$, so 5 is not a factor
    - $18 \div 6$ would be next, but we have already found that 6 was a factor
    - So we have now found all the factors of 18: 1, 2, 3, 6, 9

Help me understand

How do I find factors without a calculator?

- Use a **divisibility test**
  - Some tests are easier to remember, and more useful, than others

- Once you know that the number has a particular factor, you can divide by that factor to find the factor pair
- Instead of a divisibility test, you could use a formal written method to divide by a value
  - If the result is an integer; you have found a factor

What are some useful divisibility tests?

- A number is **divisible by 2** if the last digit is even (a multiple of 2)

- A number is **divisible by 3** if the sum of the digits is divisible by 3 (a multiple of 3)
  - 123
    $1 + 2 + 3 = 6$; 6 is a multiple of 3, so 123 is divisible by 3
  - 134
    $1 + 3 + 4 = 8$; 8 is not a multiple of 3, so 134 is not divisible by 3

- A number is **divisible by 4** if halving the number **twice** results in an integer
- A number is **divisible by 8** if it can be halved **3 times** and the result is an integer

- A number is **divisible by 5** if the last digit is a 0 or 5
- A number is **divisible by 10** if the last digit is a 0

What are prime numbers?

- A **prime** number is a number which has exactly two (**distinct**) **factors;** itself and 1
  - You should remember at least the first ten prime numbers:
    - 2, 3, 5, 7, 11, 13, 17, 19, 23, 29

- 1 only has one factor

- 2 is the only even prime number
- If a number has any factors other than itself and 1, it is not a prime number

Worked Example

Show that 51 is **not** a prime number.

If we can find a factor of 51 (that is not 1 or 51), this will prove it is not prime

51 is not even so is not divisible by 2
Next use the divisibility test for 3

5 + 1 = 6; 6 is divisible by 3; therefore 51 is divisible by 3
51 ÷ 3 = 17

The factors of 51 are 1, 3, 17 and 51

51 is not prime as it has more than two (distinct) factors

What are square numbers?

- A **square** number is the result of multiplying a number by itself
  - The first square number is $1 \times 1 = 1$, the second is $2 \times 2 = 4$ and so on

- The first 15 square numbers are: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225
  - Aim to remember at least the first fifteen square numbers

- In algebra, square numbers can be written using a power of 2
  - $a \times a = a^2$

What are cube numbers?

- A **cube** number is the result of multiplying a number by itself, twice
  - The first cube number is $1 \times 1 \times 1 = 1$, the second is $2 \times 2 \times 2 = 8$ and so on

- The first 5 cube numbers are 1, 8, 27, 64 and 125
  - Aim to remember at least the first five cube numbers
  - You should also remember $10^3 = 1000$

- In algebra, cube numbers can be written using a power of 3
  - $a \times a \times a = a^3$

What are square roots?

- The **square root** of a value, is the number that when multiplied by itself equals that value

        - e.g. The square root of 49 would be written as $\sqrt{49}$
    - Square roots can be **positive and negative**
        - e.g. The square roots of 25 are 5 and -5
    - If a negative square root is required then a - sign would be used
        - e.g. $\sqrt{25} = 5$ but $-\sqrt{25} = -5$
        - Sometimes both positive and negative square roots are of interest and would be indicated by $\pm\sqrt{25}$
- The square root of a non-square **integer** is also called a **surd**
    - e.g. $\sqrt{3}$ is a surd, as 3 is not a square number
    - Surds are **irrational** numbers
    - $\sqrt{64}$ is **rational,** as it is equal to 8
    - However, $\sqrt{2}$ is **irrational**, as 2 is not a square number
- You should aim to remember the square roots of the first 15 square numbers
    - $\sqrt{1}, \sqrt{4}, \sqrt{9}, \sqrt{16}, \sqrt{25}, \sqrt{36}, \sqrt{49}, \sqrt{64}, \sqrt{81}, \sqrt{100},$

What are cube roots?

- The **cube root** of a value, is the number that when multiplied by itself twice equals that value
    - For example, 3 is the cube root of 27
    - It is the opposite of cubing
    - Cube roots are indicated by the symbol $\sqrt[3]{}$
        - e.g. The cube root of 64 would be written as $\sqrt[3]{64}$
    - You should remember the values of the following cube roots:
        - $\sqrt[3]{1}, \sqrt[3]{8}, \sqrt[3]{27}, \sqrt[3]{64}, \sqrt[3]{125}, \sqrt[3]{1000}$

Worked Example

Write down a number which is both a cube number and a square number, and hence express this number in two different ways using index notation.

Listing the first 12 square numbers

1, 4, 9, 16, 25, 36, 49, **64**, 81, 100, 121, 144

Listing the first 5 cube numbers

1, 8, 27, **64**, 125

64 appears in both lists, it is the 8th square number and 4th cube number

64 is both a square and cube number
$64 = 8^2$ and $64 = 4^3$

- The **reciprocal** of a number is the result of **dividing 1 by that number**
  - Any number multiplied by its reciprocal will be equal to 1
- The reciprocal of 3 is $\frac{1}{3}$
  - The reciprocal of $\frac{1}{3}$ is 3
  - $3 \times \frac{1}{3} = \frac{1}{3} \times 3 = 1$
- The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$
  - The reciprocal of $\frac{3}{2}$ is $\frac{2}{3}$
  - $\frac{2}{3} \times \frac{3}{2} = \frac{3}{2} \times \frac{2}{3} = 1$
- Algebraically the reciprocal of a is $\frac{1}{a}$
  - The reciprocal of $\frac{1}{a}$ is a
  - This can also be written using a power of -1
    - $\frac{1}{a} = a^{-1}$

Worked Example

Write down a fraction that completes this calculation:

$\frac{3}{7} \times \frac{\dots}{\dots} = 1$

Recall that a number multiplied by its reciprocal is equal to 1

$$\frac{3}{7} \times \frac{\mathbf{7}}{\mathbf{3}} = 1$$

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